

System of Particles and Rotational Motion

Complete chapter notes — concepts, every derivation worked in full, the moment-of-inertia table, the linear→rotational analogy, worked examples and the traps that cost marks.

The spine of this chapter: a rigid body has two independent kinds of motion — the **translation of its centre of mass**, and **rotation about that centre of mass**. Chapter 4 handled the first. This chapter builds an exact parallel set of tools for the second: torque replaces force, moment of inertia replaces mass, angular momentum replaces momentum. Once you see the analogy, half the chapter is already learnt.

READ THIS FIRST

Most students think this chapter is "centre of mass". It is not. Centre of mass is roughly the first quarter. The bulk of the marks live in **torque, angular momentum and moment of inertia** — sections 4 to 8 below. Budget your time accordingly.

1 • Centre of mass

The centre of mass of a system is the point that moves as though the entire mass were concentrated there and all external forces acted there. It is a weighted average of position, weighted by mass.

1.1 The formulae

System	Position of the centre of mass
Two particles on a line	$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$ — divides the join internally in the ratio $m_2 : m_1$
n particles	$\vec{R}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{1}{M} \sum m_i \vec{r}_i$
Continuous body	$\vec{R}_{cm} = \frac{1}{M} \int \vec{r} dm$

Standard bodies (uniform): rod — midpoint; ring, disc, sphere, cylinder — geometric centre; triangular lamina — centroid.

The centre of mass need not lie inside the material of the body — for a ring or a horseshoe it lies in the empty space.

1.2 Centre of mass versus centre of gravity

The **centre of gravity** is the point where the total weight of the body appears to act. When g is uniform over the body — which it is for anything of ordinary size — the centre of gravity

coincides with the centre of mass. They separate only for objects large enough that g varies across them.

DERIVATION 1 Centre of mass of a uniform rod

1. Rod of mass M and length L , uniform, so the linear mass density is $\lambda = M/L$. Put the origin at one end with x measured along the rod.
2. Take a small element of length dx at distance x from the origin. Its mass is $dm = \lambda dx$.
3.
$$x_{cm} = \frac{1}{M} \int_0^L x \lambda dx = \frac{\lambda}{M} \left[\frac{x^2}{2} \right]_0^L = \frac{\lambda}{M} \cdot \frac{L^2}{2}.$$
4. Substitute $\lambda = M/L$: $x_{cm} = \frac{M}{L} \cdot \frac{1}{M} \cdot \frac{L^2}{2}.$

$x_{cm} = \frac{L}{2}$ — **the midpoint, exactly as symmetry demands.**

2 · Motion of the centre of mass

DERIVATION 2 $M\vec{a}_{cm} = \vec{F}_{ext}$

1. By definition, $M\vec{R}_{cm} = \sum m_i \vec{r}_i$, where $M = \sum m_i$ is constant.
2. Differentiate once with respect to time: $M\vec{v}_{cm} = \sum m_i \vec{v}_i = \sum \vec{p}_i = \vec{P}_{total}$.
So the total momentum of any system equals $M\vec{v}_{cm}$.
3. Differentiate again: $M\vec{a}_{cm} = \sum m_i \vec{a}_i = \sum \vec{F}_i$ — the sum of *all* forces on *all* particles.
4. Split that sum into internal and external forces. Internal forces occur in Newton-third-law pairs, $\vec{f}_{ij} = -\vec{f}_{ji}$, and therefore cancel exactly in the sum.
5. $\Rightarrow M\vec{a}_{cm} = \vec{F}_{ext}$ — the centre of mass moves exactly like a single particle of mass M acted on by the total external force.
6. If $\vec{F}_{ext} = 0$: $\vec{a}_{cm} = 0$, so $\vec{v}_{cm}$ is constant and $\vec{P}_{total}$ is conserved.

$M\vec{a}_{cm} = \vec{F}_{ext} \quad \cdot \quad \vec{F}_{ext} = 0 \Rightarrow \vec{v}_{cm} \text{ **constant**, } \vec{P}_{total} \text{ **conserved**}.$

TWO QUESTIONS THIS ANSWERS DIRECTLY

(a) A firecracker following a parabolic path explodes in mid-air. What path do the fragments' centre of mass follow? The **same parabola**. The explosion is caused entirely by *internal* forces, which cancel in pairs. The only external force is still gravity, so $\vec{a}_{cm}$ is unchanged and the centre of mass continues on its original trajectory.

(b) Passengers start walking about inside a stationary railway compartment. Does the centre of mass of the whole system move? No. The forces the passengers exert are internal to the compartment-plus-passengers system. With no external horizontal force, $\vec{v}_{cm}$ stays zero and the centre of mass remains exactly where it was — even though the compartment itself may shift slightly.

3 · The vector product — the tool the rest of the chapter runs on

$\vec{A} \times \vec{B}$ has magnitude $AB \sin \theta$ and direction perpendicular to both, given by the **right-hand rule**.

- Anticommutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$. And $\vec{A} \times \vec{A} = 0$.
- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (cyclic; reverse the order and you pick up a minus sign).
- Zero if the vectors are parallel or antiparallel; maximum if perpendicular.

Determinant form — this is how you compute torque and angular momentum in the exam:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}(A_y B_z - A_z B_y) - \hat{j}(A_x B_z - A_z B_x) + \hat{k}(A_x B_y - A_y B_x)$$

Watch the middle sign. The $\hat{j}$ term carries a minus sign in front of its determinant. That single sign is the most common error in every vector-product question.

4 · Torque — the rotational analogue of force

$\vec{\tau} = \vec{r} \times \vec{F}$, $|\vec{\tau}| = rF \sin \theta = F \times (\text{perpendicular distance from the axis})$. Unit: N m. It is a **vector**, directed along the axis of rotation.

Torque measures the turning effect of a force. Three things control it: the size of the force, how far from the axis it is applied, and the angle at which it is applied.

- **Torque is zero** when the force passes through the axis ($\theta = 0$ or 180°), however large the force is.
- **Torque is maximum** when the force is applied perpendicular to $\vec{r}$, i.e. $\theta = 90^\circ$.

FREQUENTLY ASKED

"Why are door handles fixed near the outer edge, far from the hinges?" Because $\tau = rF \sin \theta$: maximising r maximises the torque for a given force, so the door opens with the least effort. Applying the same force near the hinge gives a tiny r and almost no turning effect.

DERIVATION 3 Torque in Cartesian form

1. Write the position vector and force in components:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad \text{and} \quad \vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}.$$

$$2. \quad \vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

3. Expand along the first row, remembering the minus sign on $\hat{j}$:

$$\vec{\tau} = \hat{i}(yF_z - zF_y) + \hat{j}(zF_x - xF_z) + \hat{k}(xF_y - yF_x)$$

WORKED EXAMPLE — TORQUE ABOUT THE ORIGIN

Find the torque of the force $\vec{F} = 5\hat{i} - 2\hat{j} + 3\hat{k}$ acting on a particle whose position vector is $\vec{r} = \hat{i} - \hat{j} + \hat{k}$.

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 5 & -2 & 3 \end{vmatrix}$$

$$\hat{i}: (-1)(3) - (1)(-2) = -3 + 2 = -1$$

$$\hat{j}: -[(1)(3) - (1)(5)] = -(3 - 5) = +2$$

$$\hat{k}: (1)(-2) - (-1)(5) = -2 + 5 = +3$$

$$\vec{\tau} = -\hat{i} + 2\hat{j} + 3\hat{k} \text{ (N m)}$$

5 · Angular momentum

$\vec{L} = \vec{r} \times \vec{p}$, $|\vec{L}| = rp \sin \theta$. Unit: $\text{kg m}^2 \text{s}^{-1} = \text{J s}$. For a rigid body rotating about a fixed axis,
 $L = I\omega$.

Angular momentum is the "moment of linear momentum" — the rotational counterpart of $\vec{p}$, exactly as torque is the counterpart of $\vec{F}$.

DERIVATION 4 $\vec{\tau} = d\vec{L}/dt$, and conservation of angular momentum

1. For a single particle, $\vec{L} = \vec{r} \times \vec{p}$.
2. Differentiate using the product rule for cross products:

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$
3. **First term vanishes.** $\frac{d\vec{r}}{dt} = \vec{v}$ and $\vec{p} = m\vec{v}$, so this is $\vec{v} \times m\vec{v}$ — the cross product of two parallel vectors, which is zero.
4. **Second term.** By Newton's second law $\frac{d\vec{p}}{dt} = \vec{F}$, so $\vec{r} \times \vec{F} = \vec{\tau}$.
5. $\Rightarrow \vec{\tau} = \frac{d\vec{L}}{dt}$. Summing over a whole system, internal torques cancel in third-law pairs, leaving $\vec{\tau}_{\text{ext}} = \frac{d\vec{L}_{\text{total}}}{dt}$.
6. If $\vec{\tau}_{\text{ext}} = 0$, then $\frac{d\vec{L}}{dt} = 0$, so $\vec{L}$ is constant: $I_1\omega_1 = I_2\omega_2$.

$\vec{\tau}_{\text{ext}} = d\vec{L}/dt$ — the rotational form of Newton's second law.

$\vec{\tau}_{\text{ext}} = 0 \Rightarrow \vec{L}$ conserved $\Rightarrow I_1\omega_1 = I_2\omega_2$

5.1 Applications of angular momentum conservation

- **The spinning skater.** Pulling the arms in reduces I , so ω must rise to keep $L = I\omega$ fixed. Same for a diver tucking, and for a person on a rotating stool pulling dumbbells inward.
- **Important subtlety:** $K_{\text{rot}} = L^2/2I$, so when I falls at constant L , the kinetic energy **increases**. Energy is *not* conserved here — the extra energy comes from the muscular work done in pulling the arms in.
- **Planets.** Gravity is a central force, so it exerts no torque about the Sun; L is conserved, which is exactly Kepler's second law.

FREQUENTLY ASKED

"With no external torque on a rotating body, can the angular acceleration or the angular speed change?" **Both can.** If the body's internal mass distribution changes, I changes, so ω must change to keep $I\omega$ constant — and a changing ω is an angular acceleration. What stays fixed is L , not ω .

WORKED EXAMPLE — ANGULAR MOMENTUM FROM $\vec{r}$ AND $\vec{p}$

A particle has $\vec{r} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{p} = 3\hat{i} + 4\hat{j} - 2\hat{k}$. Show that $\vec{L}$ is perpendicular to the x -axis.

The x -component of $\vec{L} = \vec{r} \times \vec{p}$ is $L_x = r_y p_z - r_z p_y = (2)(-2) - (-1)(4) = -4 + 4 = 0$.

Since $L_x = 0$, $\vec{L}$ has no component along x — it lies in the yz -plane and is perpendicular to the x -axis.

6 · Moment of inertia

$I = \sum m_i r_i^2 = \int r^2 dm$ — where r is the **perpendicular distance from the axis of rotation**. Unit: kg m^2 .

Moment of inertia is the rotational analogue of mass: it measures a body's resistance to a change in its rotational motion. But unlike mass, **it is not a fixed property of the body** — it depends entirely on which axis you choose. Always state the axis.

It depends on three things: the total mass, the way that mass is distributed, and the position and orientation of the axis. Mass further from the axis contributes far more, because of the r^2 .

6.1 Radius of gyration

$I = MK^2 \Rightarrow K = \sqrt{I/M}$ — the **radius of gyration** K is the distance from the axis at which the whole mass could be concentrated without changing I . Unit: metre.

6.2 The standard table — learn this

Body	Axis	I	K
Thin ring / hoop	Through centre, $\perp$ to plane	MR^2	R
Thin ring	Any diameter	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Circular disc	Through centre, $\perp$ to plane	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Circular disc	Any diameter	$\frac{1}{4}MR^2$	$R/2$
Thin rod	$\perp$ through centre	$\frac{1}{12}ML^2$	$L/\sqrt{12}$
Thin rod	$\perp$ through one end	$\frac{1}{3}ML^2$	$L/\sqrt{3}$
Solid sphere	Any diameter	$\frac{2}{5}MR^2$	$R\sqrt{2/5}$
Hollow sphere (shell)	Any diameter	$\frac{2}{3}MR^2$	$R\sqrt{2/3}$
Solid cylinder	Own axis	$\frac{1}{2}MR^2$	$R/\sqrt{2}$
Hollow cylinder (thin)	Own axis	MR^2	R
Solid cylinder	$\perp$ through centre	$M\left(\frac{L^2}{12} + \frac{R^2}{4}\right)$	—

Memory aid: for the same M and R , I is largest when the mass sits furthest from the axis — **ring** > **hollow sphere** > **disc** > **solid sphere** ($1 > \frac{2}{3} > \frac{1}{2} > \frac{2}{5}$).

6.3 Rotational kinetic energy

$$K_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$$

The second form is the one exam questions exploit. **For two bodies with equal angular momenta, the one with the larger I has the smaller kinetic energy**, since $K \propto 1/I$.

7 · Rotational dynamics and kinematics

$$\tau = I\alpha \text{ — the rotational } F = ma. \quad W = \tau\theta \quad \cdot \quad P = \tau\omega \quad \cdot \quad \text{angular impulse } \tau\Delta t = \Delta L$$

7.1 Rotational kinematic equations (constant α)

Identical in form to the linear ones — just swap the symbols:

$$\omega = \omega_0 + \alpha t \quad \cdot \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \cdot \quad \omega^2 = \omega_0^2 + 2\alpha\theta \quad \cdot \quad \theta_{n^{\text{th}}} = \omega_0 + \frac{\alpha}{2}(2n - 1)$$

Linking linear and angular quantities: $v = \omega r$, $a_t = \alpha r$, $a_c = \omega^2 r$, $s = r\theta$.

8 · Equilibrium of a rigid body

A rigid body can both translate and rotate, so it needs **two** conditions to be in equilibrium — this is the key difference from a particle:

$$\sum \vec{F} = 0 \text{ (no translation) } \text{ AND } \sum \vec{\tau} = 0 \text{ (no rotation)}$$

Both are needed. A couple has zero net force yet still spins the body, so $\sum F = 0$ alone is not enough.

8.1 Principle of moments

In rotational equilibrium, the sum of the anticlockwise moments equals the sum of the clockwise moments: $F_1 d_1 = F_2 d_2$.

This is the theory of the lever. The **mechanical advantage** is

$$\text{MA} = \frac{\text{load}}{\text{effort}} = \frac{d_{\text{effort}}}{d_{\text{load}}}$$

— so a long effort arm lets a small effort balance a large load.

8.2 Couple

A **couple** is a pair of forces that are equal in magnitude, opposite in direction, and **not collinear** (they act along different lines).

- Net force = 0, so a couple produces **no translation**.
- Net torque = $F \times d$, where d is the perpendicular distance between the lines of action.
- **The torque of a couple is the same about every point** — you can take moments anywhere and get the same answer.
- It produces **pure rotation**: turning a tap, a steering wheel, a screwdriver, winding a clock.

WORKED EXAMPLE — A NON-UNIFORM BAR ON TWO STRINGS

A non-uniform bar of weight W and length 2 m hangs at rest from two strings making angles of 36.9° and 53.1° with the vertical. Find the distance of its centre of gravity from the left end. (Use $\cos 36.9^\circ = 0.84$, $\sin 36.9^\circ = 0.55$, $\cos 53.1^\circ = 0.68$, $\sin 53.1^\circ = 0.74$.)

Step 1 — horizontal forces balance.

$$T_1 \sin 36.9^\circ = T_2 \sin 53.1^\circ \Rightarrow 0.55 T_1 = 0.74 T_2 \Rightarrow T_1 = \frac{0.74}{0.55} T_2$$

Step 2 — vertical forces balance.

$$T_1 \cos 36.9^\circ + T_2 \cos 53.1^\circ = W$$

$$\left(\frac{0.74}{0.55} T_2\right) (0.84) + 0.68 T_2 = W \Rightarrow T_2(1.130 + 0.68) = W \Rightarrow T_2 \approx 0.552 W$$

Step 3 — take torques about the left end, which eliminates T_1 entirely (its line of action passes through that point). The weight acts downward at distance x ; the vertical component of T_2 acts upward at 2 m.

$$W \times x = (T_2 \cos 53.1^\circ) \times 2 = (0.552W)(0.68)(2)$$

$x \approx 0.75$ m from the left end

The technique to carry away: take moments about a point through which an unknown force acts. That force then contributes zero torque and drops out of the equation.

9 · The linear ↔ rotational analogy

This table is worth several marks on its own, and it also means you never have to learn the rotational formulae separately — you derive them from the linear ones by substitution.

Linear motion	Rotational motion (fixed axis)
Displacement s	Angular displacement θ
Velocity $v = ds/dt$	Angular velocity $\omega = d\theta/dt$
Acceleration $a = dv/dt$	Angular acceleration $\alpha = d\omega/dt$
Mass m (inertia)	Moment of inertia I
Force F	Torque $\tau = rF \sin \theta$
$F = ma$	$\tau = I\alpha$
Momentum $p = mv$	Angular momentum $L = I\omega$
$F = dp/dt$	$\tau = dL/dt$
Impulse $F\Delta t = \Delta p$	Angular impulse $\tau\Delta t = \Delta L$
Work $W = Fs$	$W = \tau\theta$
KE $= \frac{1}{2}mv^2 = p^2/2m$	KE $= \frac{1}{2}I\omega^2 = L^2/2I$
Power $P = Fv$	$P = \tau\omega$
$v = u + at$	$\omega = \omega_0 + \alpha t$
$s = ut + \frac{1}{2}at^2$	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

DERIVATION 5 Angular momentum of two particles is independent of the origin

Two particles, each of mass m and speed v , travel in opposite directions along parallel lines separated by a distance d . Show that the total angular momentum is the same about every point.

1. Choose any origin O . Let particle 1 be at $\vec{r}_1$ moving with velocity $\vec{v}$, and particle 2 at $\vec{r}_2$ moving with $-\vec{v}$.
2. Total angular momentum: $\vec{L} = \vec{r}_1 \times (m\vec{v}) + \vec{r}_2 \times (-m\vec{v})$.
3. Factor out: $\vec{L} = m(\vec{r}_1 - \vec{r}_2) \times \vec{v}$.
4. But $(\vec{r}_1 - \vec{r}_2)$ is the **relative** position vector of one particle from the other — it does not involve O at all.
5. Since the choice of origin has cancelled, $\vec{L}$ is the same about every point. Its magnitude is $|\vec{L}| = mvd$, where d is the perpendicular separation of the two lines.

$\vec{L} = m(\vec{r}_1 - \vec{r}_2) \times \vec{v}$, **magnitude mvd — independent of the origin. This is the defining property of a couple applied to angular momentum.**

10 · Only if covered in class — axis theorems and rolling

These were removed in the rationalised syllabus, so check whether your teacher covered them.

- **Parallel axis theorem.** $I = I_{cm} + Md^2$, where d is the perpendicular distance between two *parallel* axes and one of them passes through the centre of mass.
Check: a rod about its end $= \frac{1}{12}ML^2 + M(L/2)^2 = \frac{1}{3}ML^2$ ✓
- **Perpendicular axis theorem (plane laminae only).** $I_z = I_x + I_y$, where x and y lie in the plane and z is perpendicular to it, all three meeting at a point.
Check: a disc $— \frac{1}{2}MR^2 = \frac{1}{4}MR^2 + \frac{1}{4}MR^2$ ✓ (Never apply this to a sphere or a cylinder.)
- **Rolling without slipping.** $v_{cm} = \omega R$; the contact point is instantaneously at rest; the topmost point moves at $2v_{cm}$.
- **Rolling kinetic energy.** $K_{total} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv_{cm}^2 \left(1 + \frac{K^2}{R^2}\right)$, where K here is the radius of gyration.
- **Rolling down an incline θ .** $a = \frac{g \sin \theta}{1 + K^2/R^2}$ and $v_{bottom} = \sqrt{\frac{2gh}{1 + K^2/R^2}}$.
Since a falls as K^2/R^2 rises, the arrival order down the same incline is **solid sphere** ($\frac{2}{5}$) → **disc** ($\frac{1}{2}$) → **hollow sphere** ($\frac{2}{3}$) → **ring** (1), independent of M and R .

TRAPS THAT COST MARKS

1. **I is not a property of the body alone — it depends on the axis.** Always name the axis before quoting a value.
2. **In $I = \sum mr^2$, r is the perpendicular distance from the AXIS** — not the distance from the centre or from the origin.
3. **When L is conserved and I decreases, ω increases AND K_{rot} increases.** Kinetic energy is not conserved — internal work supplies the difference. Saying "energy is conserved" here is wrong.
4. **The perpendicular-axis theorem applies to plane laminae only;** the parallel-axis theorem needs one axis through the centre of mass.
5. **Rigid-body equilibrium needs both $\sum F = 0$ and $\sum \tau = 0$.** Quoting only the first is a half-answer.
6. **$\vec{\omega}$, $\vec{\tau}$ and $\vec{L}$ point along the AXIS** (right-hand rule), not along the direction of motion.
7. **Zero external torque does not fix ω** — it fixes L . If I changes, ω changes.
8. **The minus sign on the $\hat{j}$ term** in every cross-product determinant. Check it every time.
9. **Zero external torque says nothing about v_{cm} .** The velocity of the centre of mass is controlled by external *force*, not torque — a favourite false assertion.

Chapter 6 of the rationalised NCERT (2023–24 onward). Every derivation above is written the way it should appear in the answer script: numbered steps, the key substitution shown explicitly, and the final result stated separately.